SARWAN KUMAR PANDEY

Lucknow, India | +91-6206805982 | sarwanpandey75@gmail.com | linkedin.com/in/sarwan-pandey

| github.com/sarwan-pandey

CAREER OBJECTIVE

B.Tech Computer Science student focused in the Artificial Intelligence and Machine Learning, with hands-on projects experience in Python, Scikit-Learn, Hugging Face, RAG, FAISS, FastAPI and Computer Vision.

TECHNICAL SKILLS

Languages: Python, SQL, C++
AI / ML: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, RAG, Reinforcement Learning
Libraries & Frameworks: TensorFlow, Keras, Scikit-learn, NumPy, Pandas, OpenCV, PyTorch, Scikit-learn
Robotics: ROS 2, Simulation, Navigation, Localization, SLAM
Tools & Deployment: FastAPI, Git, Docker

EXPERIENCE

Center of Intelligent Robotics, IIIT Allahabad | *Jun 2026 – Aug 2026*

- Applied reinforcement learning, NLP, and computer vision techniques through hands-on implementation.
- Developed **ROS 2**-based robotics workflows covering simulation, navigation, localization, **SLAM**, and vision integration.
- Integrated AI/ML models with robotic and drone systems, covering evaluation, optimization, and deployment.

PROJECTS

FaceID Studio – Real-Time Face Recognition | *Python, OpenCV, InsightFace, ONNX Runtime*

- Engineered a real-time face recognition application with **multithreaded** camera processing.
- Cached face embeddings, added configurable recognition thresholds, and implemented gallery management.

Retrieval-Augmented Generation (RAG) Pipeline | *Python, LangChain, FAISS, Hugging Face, Transformers*

- Created a knowledge-based question-answering system using document **embeddings** and **semantic similarity search** (FAISS).
- Integrated retrieval with **LLM**-based generation to improve response relevance.

Sequence Prediction using RNNs | *Python, TensorFlow, Keras, NumPy, Pandas*

- Implemented **time-series forecasting** and next-word prediction models using Embedding and SimpleRNN layers.
- Benchmarked **3 recurrent architectures** (RNN, LSTM, GRU) using Mean Squared Error (MSE).

Image Classifier Web Application | *Python, FastAPI, TensorFlow, Keras, Scikit-learn*

- Designed a **CNN** classifier for the 10-class, 60,000-image CIFAR-10 dataset.
- Deployed the TensorFlow/Keras model through a **FastAPI** service for real-time image prediction.

EDUCATION

B.Tech, Computer Science (AI & ML)

SR Institute of Management & Technology | 2023-2027

CERTIFICATIONS

- AWS Academy Graduate – Machine Learning Foundations
- AWS Academy Graduate – Generative AI Foundations
- AI/ML for Geodata Analytics – ISRO (IIRS Dehradun)